PANJAB UNIVERSITY, CHANDIGARH

(Estted. under the Panjab University Act VII of 1947- enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI

FOR

MASTERS IN GEOINFORMATICS

(SEMESTER SYSTEM) 1st & 2nd Semester

(w.e.f. 2019-20)

APPLICABILITY OF REGULATIONS FOR THE TIMEBEING IN FORCE

Notwithstanding the integrated nature of a course spread over more than one academic year, the regulations in force at the time a student joins a course shall hold good only for the examinations held during or at the end of the academic year. Nothing in these regulations shall be deemed to debar the University from amending the regulations subsequently and the amended regulations, if any, shall apply to all students whether old or new.

MASTERS IN GEOINFORMATICS
(Semester System)
Outlines of Course Content and Syllabi
for the w.e.f. the session 2019-20

ABOUT THE COURSE

Masters in Geoinformatics is a multi-disciplinary & professional postgraduate course offered by Department of Geography, Panjab University Chandigarh from the academic session 2019-20. Department of Geography, Panjab University Chandigarh has been a pioneer in introducing such professional courses on geospatial technology. We first introduced a PG Diploma course in Remote Sensing and Geographic Information Systems in 2001 which was later upgraded into two year Masters in Remote Sensing and Geographic Information Systems (RS & GIS).Keeping pace with changing technological scenario in the field, the course was restructured and renamed as Masters in Geoinformatics which is now more robust in terms of course content and professional attitude as well as more flexible and interdisciplinary in nature. This two year course, divided into four semesters, is aimed at developing scientific, technical and professional attitude and skills among students.

WHY STUDY WITH US

Department of Geography, Panjab University offers a postgraduate Masters course that aims at developing technically skilled and scientifically oriented professionals in the field of geoinformatics. The course is dynamic, uniquely designed and flexible that not only produce highly trained professionals in the field of geospatial technology but also encourages innovation and out-of-box thinking. During the two year stay with us students are exposed to a wide range of knowledge learning and skill building practices. We offer a wide spectrum of learning skills ranging over conceptual and theoretical clarity of the subject, technical capabilities, analytical and decision making qualities. Using a focussed field based approach, we prepare our students to develop practical understanding of geospatial processes, mechanisms and operational aspects of space based technology to seek effective solutions to problems. ***Some of the highlights of our curriculum are:***

- Conceptual & Theoretical Clarity of Key Concepts in Geoinformatics.
- Exposure to Remote Sensing and Digital Image Processing.
- Technical skills in the field of Photogrammetry.
- Innovative learning of GIS and GNSS.
- Understanding of basic Computer Skills and Programming.
- Development of Analytical and Creative Skills in Spatial Analysis.
- Specialization in Geosciences, Urban & Regional Planning and Disaster Management.
- Upgradation of Skills for Field-work Based Data Collection of Geo-Information.
- Improvement in Quantitative and Qualitative Research Skills.
- Project Formulation and Management.
- Report Writing and Delivering High Quality Output.

GUIDELINES FOR CONTINUOUS INTERNAL ASSESSMENT (20%)

For Regular Students of Masters in Geoinformatics (Semester System)
(Effective from the First Year Admissions for the Academic Session 2019-2020)

1. The Syndicate has approved the following guidelines, mode of testing and evaluation including Continuous Internal Assessment of Students:

i. Terminal Evaluation : **80%**(Theory 50% and Practical 30% as specified in syllabus)
ii. Continuous Assessment : **20%**
iii. Continuous Assessment may include written assignment, snap tests, participation in discussions in the class, term papers, attendance etc.
iv. In order to incorporate an element of Continuous Internal Assessment of students, the colleges/ Departments will conduct one written test as quantified below:

a) Written Test	: 25 (reduced to 5)
b) Snap Test	: 25 (reduced to 5)
c) Term Paper	: 25 (reduced to 5)
d) Participation in class discussions	: 15 (reduced to 3)
e) Attendance	: 10 (reduced to 2)
	Total: 100 reduced to 20

2. Weightage of 2 marks for attendance component out of 20 marks for Continuous Assessment shall be available only to those students who attend 75% and more of classroom lecture/ seminars/ workshops. The break-up of marks for **attendance component** for theory papers shall be as under:

Attendance Component	*Marks for Theory Papers*
(a) 75% and above upto 85%	**: 1**
(b) Above 85%	**: 2**

3. It shall **not be compulsory** to pass in Continuous Internal Assessment. Thus, whatever marks are secured by a student out of 20% marks, will be carried forward and added to his/her score out of 80%, i.e. the remaining marks allocated to the particular subject and, thus, he/she shall have to secure pass marks both in the University examinations as well as total of Internal Continuous Assessment and University examination.

4. Continuous Internal Assessment awards from the affiliated Colleges/Departments must be sent to the Controller of Examinations, by name, **two weeks before** the commencement of the particular examination on the *proforma* obtainable from the Examination Branch.

Masters in Geoinformatics COURSE OUTLINE			
Course Code	**Course Title**	**Max. Marks**	**Credits**
SEMESTER-I			
GI 101	Fundamentals of Remote Sensing	100	4
GI 102	Fundamentals of Photogrammetry	100	4
GI 103	Fundamentals of Geographic Information Systems	100	4
GI 104	Fundaments of Computer and Computer Programming	100	4
	Semester Total	*400*	*16*
SEMESTER-II			
GI 201	Cartography	100	4
GI 202	Spatial Analysis	100	4
GI 203	Digital Image Processing	100	4
GI 204	Fundaments of Global Positioning System	100	4
	Semester Total	*400*	*16*
SEMESTER-III			
GI 301	Research Methodology and Internship Report	100	4
GI 302	Geoinformatics for Geosciences	100	4
GI 303	Geoinformatics for Urban and Regional Planning	100	4
GI 304	Geoinformatics for Disaster Management	100	4
	Semester Total	*400*	*16*
SEMESTER-IV			
GI 401	Project Report 1. Formulation of Project Proposal 2. Mid Term appraisal 3. Pre-submission Presentation 4. Report Evaluation 5. Presentation 6. Viva-voce	 50 50 50 150 50 50	16
	Semester Total	*400*	*16*
	GRAND TOTAL (Semester-I, II, III, IV)	**1600**	**64**

NOTE:

1. After successful completion of first two semesters of the Masters course, there shall be a provision for the candidate/s to opt out of the Course. In such an instance S/He will be awarded Diploma in Geoinformatics.
2. The minimum marks required to pass the examination shall be 50% in aggregate in each semester and 45% in each individual paper: (i) theory paper, (ii) practical examination, (iii) internship report and (iv) project report.
3. The Internship Report shall be prepared by the candidate on the basis of work done or experience gained on visit(s) to Remote Sensing Centre(s)/Institute (s) in India. The candidate shall submit three copies of his/her Internship Report ten days before the commencement of the theory examination of the said semester.
4. In case of Project Report, the Board of Control shall assign Supervisor/Co-supervisors for each candidate for supervising candidate's Project Report on an approved topic. External Supervisor/Co-supervisors may be allotted, subject to their eligibility to be determined by the Board of Control.

SEMESTER-I

GI 101: FUNDAMENTALS OF REMOTE SENSING

Max. Marks: 100
Theory: 50
Practical: 30
Internal Assessment: 20

Objectives: The aim of this course is to introduce the students with basic principles of remote sensing, satellite systems, their functioning and applications of space based technology. It also intends to expose students to technical skills required to analyse and interpret satellite data.

(A) THEORY **Marks: 50**

Unit-I

Fundamental Concepts of Remote Sensing:

i. Remote Sensing: Concept, Types and Applications
ii. Remote Sensing Platforms, Sensors and Scanning Systems
iii. Major Satellite Systems, their Sensors and Applications

Unit-II

Energy Principles and Interaction Mechanisms:

iv. Radiation Principles; Electromagnetic Spectrum
v. Energy-Atmosphere Interaction; Atmospheric Windows
vi. Energy-Earth Interaction; Spectral Signatures of Surface Features

Unit-III

Image Processing and Interpretation:

vii. Image: Meaning, Types and Characteristics
viii. Resolution: Spatial, Spectral, Radiometric and Temporal
ix. Basics of Image Processing; Elements of Image Interpretation.

Unit-IV

Advances in Remote Sensing :

x. Thermal Remote Sensing: Concept, Mechanism and Application
xi. Microwave Remote Sensing: Concept, Mechanism and Application
xii. Hyperspectral Remote Sensing: Concept, Mechanism and Application

Note:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Introduction to Wavelength, Frequency and Spectrum
2. Calculation of Frequency, Wavelength, Energy and Radiant Exitance
3. Comparing Satellite Imageries of Various Spatial Resolution
4. Spectral Signature Analysis of Surface Features
5. Visual Interpretation of Features from Satellite Image
6. Introduction to Image Processing Software
7. Polynomial Rectification: Georeferencing Satellite Image
8. Preparing Image Subset, Mosaic, Layer stacking and Resolution merging
9. Digital Image Interpretation and Analysis:
   - Band Combination, Histogram and Pixel Value Analysis
   - Identification and interpretation of Features
10. Digital Image Classification-I: Supervised Classification

11. Digital Image Classification-II: Unsupervised Classification
12. Map Composition and Layout

Note: The practical examination shall be conducted by a team of three examiners, including the faculty teaching the paper, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Campbell, J. B. and R.H. Wynne, *Introduction to Remote Sensing*, Guilford Press, 2012.
2. Jensen, J.R., *Introductory Digital Image Processing: A Remote Sensing Perspective*, Pearson Prentice Hall, 2005.
3. Joseph, G. *Fundamentals of Remote Sensing*, Second Edition, Universities Press, 2005.
4. Lillesand, T.M., R.W. Kiefer and J.W. Chipman, *Remote Sensing and Image Interpretation*, 5th Edition, John Wiley and Sons, 2004.
5. Mather, P.M. and M.G. Koch, *Computer Processing of Remotely-Sensed Images: An Introduction,* John Wiley & Sons, 2011.
6. Schowengerdt, Robert A. *Remote Sensing: Models and Methods for Image Processing*, Second Edition, Academic Press, San Diego, 2017.
7. Srivastava, G.S., *An Introduction to Geoinformatics*, McGraw Hill Education, India, New Delhi, 2014

Further Readings:

1. Gomarasca, Mario A, *Basics of Geomatics*, Springer, Heidelberg, 2009.
2. Dong, P. and Qi Chen, LiDAR Remote Sensing and Applications. CRC Press, 2018.
3. Nag, P. and M. Kudrat, *Digital Remote Sensing*, Concept Pub. Company, New Delhi, 1998.
4. Njoku, E.G., *Encyclopaedia of Remote Sensing*, Springer, New York, 2014
5. Rees, W.G., *Physical Principles of Remote Sensing,* 3rd Ed., Cambridge University Press, 2012.
6. Richards, John A. *Remote Sensing Digital Image Analysis: An Introduction,* 5th Edition, Springer-Verlag Berlin Heidelberg, 2013.

Pedagogy: Students may be taken to field for verification of identifiable features on satellite imageries; they may be asked to trace the temporal and spatial changes. Students may be taken to different institutions to acquaint with different remote sensing techniques. Professionals may also be invited to interact and deliver special lectures to students on remote sensing techniques.

GI 102: FUNDAMENTALS OF PHOTOGRAMMETRY

Max. Marks : 100
Theory : 50
Practical : 30
Internal Assessment : 20

Objectives: To provide a general overview of Photogrammetry, its theory and general working principles. It introduces the basic concepts and significance of Photogrammetry. Further, a practical understanding of principles of measurement fromvertical aerial photographs is provided.

(A) THEORY **Marks: 50**

Unit-I

i. Photogrammetry: Definition and Types
ii. Historical Developments of Photogrammetry
iii. Flight Planning Parameters for Photogrammetry
iv. Basic Elements of Overlap and Side-overlap, Crab and Drift

Unit-II

v. Basic Geometry of Aerial Photographs
vi. Calculation of Photo Scale
vii. Relief Displacement: Concept and Calculations
viii. Parallax: Concept and Determination
ix. Calculation of Height of Objects on Vertical Aerial Photograph

Unit-III

x. Stereoscopy: Definition
xi. Principles of vision, Binocular Vision, Stereoscopic Vision Testing,Pseudoscopic Views
xii. Types of Stereoscopes and their Operations

Unit-IV

xiii. Planimetric Positions
xiv. Photogrammetry in Space Age
xv. Applications of Photogrammetry

Note:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from eachunit. The candidates will attempt 4 questions selecting one from each unit carrying 9marks each (total 36 marks). These will be in addition to the Compulsory Question atserial number I.

(B) PRACTICAL **Marks: 30**

1. Distinction between Map, Photo and Stereoscopic view.
2. Arrangement of Aerial Photographs with the help of numbers given on the photograph and preparation of Photo Index.
3. Understanding the simple geometry of aerial photographs
4. Interpretation of information given on the margins of photograph and satellite imageries.
5. Locating principal point, conjugate principal point, flight line
6. Stereoscopic vision Test, selection of stereo pair and its orientation and placing under the stereoscope for stereovision.
7. Identification and mapping of features from single photograph.
8. Identification and mapping of features from a stereo pair and their verification in the field.
9. Measurement of scale, height and slope from vertical aerial photographs.
10. Measurement of relief displacement and parallax from vertical aerial photographs.
11. Identification of planimetric positions

Note: The practical examination shall be conducted by a team of three examiners, including the faculty teaching the paper, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. McGlone, J. Chris, *Manual of Photogrammetry (Sixth Edition),* American Society ofPhotogrammetry and Remote Sensing, 2013.
2. Mikhail, Edward M. et al, *Introduction to Modern Photogrammetry*, John Wiley &Sons Inc. 2001.
3. Moffitt, F. H., *Photogrammetry*, 3rd Edition, Harper & Row, NY, 1980.

4. Paul Wolf et al., *Elements of Photogrammetry with Application in GIS*, Fourth Edition,McGraw-Hill Professional, 2013.
5. Srivastava, G.S., *An Introduction to Geoinformatics*, McGraw Hill Education, India, New Delhi, 2014
6. Wolf, P. R., *Elements of Photogrammetry*. McGraw-Hill, NY, 1983.
7. Zorn, H. C., *Introductory Course in Photogrammetry*. 6th Ed. ITC, Netherlands, 1980.

Further Reading:

1. Burnside, C. D., *Mapping from Aerial Photography,* 2nd Ed, Collins, 1985.
2. Campbell, J.B., *Introduction to Remote Sensing*, 3rd ed., The Guilford Press, 2002.
3. Curran Paul, J., *Principles of Remote Sensing*, UK: ELBS, 1984.
4. Gomarasca, Mario A, *Basics of Geomatics*, Springer, Heidelberg, 2009.
5. Joseph, George, *Fundamentals of Remote Sensing,* Universities Press India, 2007.
6. Lillesand, T. M.; R.W. Kiefer, and J.W. Chipman, *Remote Sensing and Image Interpretation*, 5th Edition, Wiley, 2007.
7. Sabins, Floyd F., *Remote Sensing: Principles and Interpretation,* New York: WHFreeman and Company, 2007.

Pedagogy: Students will be demonstrated the difference between qualitative andquantitative understanding of various aspects of Photogrammetry. Demo exercise onmeasurements on the map and photo will make students aware the importance ofPhotogrammetry in objective and precise mapping using aerial photographs.

GI 103: FUNDAMENTALS OF GEOGRAPHIC INFORMATION SYSTEMS

Max. Marks : 100
Theory : 50
Practical : 30
Internal Assessment : 20

Objectives: The main objective of this course is to expose the students to fundamental principles of Geographical Information Systems including basic concepts and definitions, methods and techniques. It also introduces basics of GIS software and tools.

(A) THEORY **Marks: 50**

Unit- I

Concepts and Definitions

i. Geographic Information Systems (GIS): Definition and Developments
ii. Spatial data base: Points, Lines and Polygons
iii. GIS Components: Hardware and Software

Unit-II

Functional Elements

iv. Data Acquisition, Input and Editing
v. Data Manipulation and Topology Creation
vi. Data Analysis and Query

Unit-III

Data Management and Structure

vii. Data Base Management System (DBMS): Purpose and structure
viii. Classification: Raster, Vector, Quadtree, Hierarchical, Network and Relational

Unit-IV

GIS Applications

ix. GIS Applications
x. GIS: Decision Support System

Note:
1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. GIS Software: Introduction to Arc GIS
2. Georeferencing Maps/Images
3. Digitization of Raster Map: Point, Line and Polygon Features
4. Building Topology
5. Preparation of Attribute Tables, Editing and Joining Tables
6. Spatial Representation: Mapping Techniques
7. Spatial Representation: Symbolizing and Map Layouts
8. Basic Analysis in GIS: Buffering, Overlay and Query Building

Note: The practical examination shall be conducted by a team of three examiners, including the faculty teaching the paper, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Fazal, S., *GIS Basics*, New Age International Publishers, New Delhi, 2008.
2. Fazal, S. and Rahman, A., *Geographic Information System (GIS) Terminology*, New Age International Publishers, New Delhi, 2007.
3. Goodchild, M.F., *Introduction to GIS*, Santa Barbara, California, & Karan K. Kemp (eds.) NGGIA, 1990.
4. Harvey, F., *A Primer of GIS: Fundamental Geographic and Cartographic Concepts*, Guilford Press, New York, 2008.
5. Heywood, I., Sarah Cornelius and Steve Carver, *An Introduction to Geographical Information Systems*, 3rd Edition, Pearson, Dorling Kindersley (India) Pvt. Ltd, 2010.
6. Kang-tsung (Karl) Chang, *Introduction to Geographic Information Systems*, 4th Edition, Tata McGraw-Hill, New Delhi, 2009.
7. Lo, C.P and Albert K.W., Yeung, *Concepts and Techniques of Geographic Information Systems*, 2nd ed., Pearson Education Inc., Toronto, Canada, 2007.
8. Longley, Paul A., Goodchild, Michael F. Maguire, David J., and David W. Rhind, *Geographic Information Systems and Science*, 3rd ed., John Wiley and Sons, England, 2011.
9. Maguire, D.J.; M. Batty and M.F. Goodchild, *GIS, Spatial Analysis and Modeling*, ESRI Press, 2005.
10. Siddiqui, M.A.,*Introduction to Geographical Information Systems*, Sharda Pustak Bhavan, Allahabad, 2006.
11. Wise, S., *GIS Fundamentals*, 2nd Edition, CRC Press, Taylor & Francis Group, 2013.

Further Readings:

1. Aronoff, S., *Geographic Information Systems: A Management Perspective*, WDL Publications Ottawa, Canada, 1992.
2. Bolstad, P.: *GIS Fundamentals: A First Text on Geographic Information Systems*, 5th ed., XanEdu Publishing Inc, 2016.

3. Brewer, C.A.,*Designing Better Maps: A Guide for GIS Users*, ESRI Press, 2005.
4. Burrough, P. A., McDonnell, R. A. and Lloyd, C.D.,*Principles of Geographical Information Systems*, 3rd. ed., Oxford University Press, Toronto, 2016.
5. DeMeers, M.N., *Fundamentals of Geographic Information Systems*, 4th Edition, John Wiley and sons, 2008.
6. ESRI : *Understanding GIS*, Environmental Systems Research Institute, U.S.A., 1993.
7. ESRI: *Esri Map Book*, Volume 32, Esri Press, 2017.
8. Graser, A.: *Learning QGIS: Create great maps and perform geoprocessing tasks with ease*, 3rd. ed., Packt Publishing, 2016.
9. Pinde Fu, P..*Getting to Know Web GIS*, Esri Press, 2018.
10. Rashid, S.M., *Remote Sensing in Geography*, Manak Publications, Delhi, 1993.
11. Shen, Zhenjiang, *Geospatial Techniques in Urban Planning*, Springer-Verlag Berlin Heidelberg, 2012.
12. Tahir, H., Abineh, T., and Ayele, B.,*Geographic Information System (Theory and Practical Manual)*, Lambert Academic Publishing, 2015.

Pedagogy: Taking students for fieldwork, inviting professionals for lectures, taking students to institutes/training centres engaged in imparting professional training and providing research consultancy in business, tourism and area/regional planning will be the main component of the teaching methodology.

GI 104: FUNDAMENTALS OF COMPUTERS AND COMPUTER PROGRAMMING

Max. Marks : 100
Theory : 50
Practical : 30
Internal Assessment : 20

Objectives: This course intends to develop basic skills and understanding of computer operations; skills of Word Processing for written communication of results and documentation of work performed. It aims to impart skills for data management, charting, and visualization; and to expose students to the effective techniques of using search engines & the services of Internet as well as imparting programming skill for developing programs in 'C' language.

(A) THEORY **Marks: 50**

Unit-I

Computer Concepts, Operating Systems and Application Software

i. Introduction to Computers, Computer System, Hardware, Software, Input and Output Devices
ii. Data Representation; Fundamental Concepts of Operating Systems.
iii. Introduction to Windows Operating System; MS-Word and MS-Excel Software.

Unit-II

Computer Graphics, Computer Networks and Web Application Development

iv. Fundamentals of Computer Graphics, Image Formats
v. Introduction to Computer Networks
vi. Overview of Internet and its services, Search Engines, Techniques to search effectively
vii. Introduction to web application development using Hypertext Markup Language (HTML)

Unit-III

Data Concepts and Information Systems

viii. Key Database Concepts; Database Management Systems, Type of Data Base Management System (DBMS)
ix. Structured Query Language (SQL); Data Visualization, Data Mining
x. Introduction to Information Systems; Types of Information Systems

Unit-IV

Programming Concepts

xi. Concept of a Program, Data Types, Keywords, Structure of a Program.
xii. Writing 'C' Programs:Statements, Types of Statements
xiii. Library Functions and User Defined Functions, Arrays, File Handling.

Note:

1. A Compulsory Question at serial number I containing 10 short answer type questionsshall be set covering the whole syllabus. Student will attempt any 7 parts in about 25- 30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Hands on experience on Windows operating system.
2. Hands on experience on MS-Word
3. Hands on experience on MS-Excel
4. Creating web pages using HTML,
5. Techniques to use search engines effectively.
6. Writing queries using SQL.
7. Writing C programs.

Note: The practical examination shall be conducted by a team of three examiners, including the faculty teaching the paper, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings

1. Basandra, S. K., *Computers Today*, Galgotia Publications Pvt. Ltd., 2010.
2. Frye, C., *Microsoft Excel 2013: Step by Step*, Microsoft Press, 2013.
3. Lambert, J. and J. Cox, *Microsoft Word 2013: Step by Step*, Eastern Economy Edition, 2013.
4. Miller, M., *Easy Computer Basics, Windows 8.1 Edition*, Pearson Education, Inc. 2014.
5. Online Resources: www.w3schools.com for HTML and SQL. (*Live and latest).*
6. Salaria, R.S. *Computer Concepts and Programming in C*, Salaria Publishing House, 2013.
7. Yang, Chaowei, *Introduction to GIS Programming and Fundamentals with Python and ArcGIS*, CRC Press, 2017.

Further Readings

1. Freedman, J., *Microsoft Word 2013: Plain & Simple*, O'Reilly Media Inc., 2013.
2. Lutz, M. *Programming Python*, 3rd Edition, O'Reilly Media, Inc. 2006.
3. Norton, P., *Introduction to Computers*, 6th Edition, McGraw-Hill Education, 2004.
4. Norton, P., *Computing Fundamentals*, Student Edition, McGraw-Hill, 2004.
5. Tateosian, L. *Python for ArcGIS*, Springer, 2015

Pedagogy: Bottom-up approach of explaining concepts; programming exercises to simulate interest to practice programming applications; Audio visual presentations; Test/Assignment as per the latest in industry/real-life. Students will be encouraged to ask questions and participate in the classroom discussion.

SEMESTER-II

GI 201: CARTOGRAPHY

Max. Marks : 100
Theory : 50
Practical : 30
Internal Assessment : 20

Objective: The course is designed to emphasize the relevance and scope of cartography in GIS. It shall highlight the value of this vital field in the current and highly versatile field of geoinformatics.

(A) THEORY **Marks: 50**

Unit-I

Basic Concepts of Cartography

i. Cartography: Definition, Scope and Significance
ii. Historical Perspective of Cartographic Development
iii. Traditional and Digital Cartography; Cartography and GIS

Unit-II

Data Types and Map Basics

iv. Levels of Measurements; Data: Types (Spatial and Attribute data), Sources and Analysis
v. Map Types and Classifications; Topographic Maps
vi. Map Projections, Datum and Coordinate System

Unit-III

Map Designs and Production

vii. Principles of Symbolization; Basics of Colour Schemes and Representation
viii. Map Lettering and Legend Presentation
ix. Map Layout and Product Generation

Unit-IV

Interpretation of Topographical Maps

x. Relief Representation: Contours, Hachures, Hill Shading
xi. Relief (Absolute and Relative) and Profiles (Longitudinal and Transverse)
xii. Slope Analysis (Wentworth and Robinson); Reading Topographical Maps

Note:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25- 30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Classification of Topographical Maps
2. Slope Analysis (Wentworth and Robinson)
3. Generation of Hill Shading Map
4. Profiles: Longitudinal and Transverse Profiles
5. Interpretation of Physical and Cultural features from Topographical Maps
6. Thematic Maps: Dot Map, Choropleth Map, Proportionate Circles, Isopleth Map

Note: The practical examination shall be conducted by a team of three examiners, including the faculty teaching the paper, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings

1. Cromley, R.G., *Digital Cartography*, Prentice-Hall of India, New Delhi, 1992.
2. Crampton, W.C., *Mapping: A Critical Introduction to Cartography and GIS,* John Wiley & Sons, New York, 2010.
3. Dent, B.D., *Cartography-Thematic Map Design*, WBC McGraw–Hill, Boston, 1999.
4. John Krygier and Denis Wood, *Making Maps: A visual guide to map design for GIS,* Guilford Publications, New York, 2005.
5. Keates, J.S., *Cartographic Design and Production*, Longman, London, 1998.
6. Misra, R.P. and Ramesh, A., *Fundamentals of Cartography*, Concept PublishingCompany, New Delhi, 1989.
7. Monkhouse, F.J., *Maps and Diagrams*, Methuen and Co., London, 1994.
8. Nag, P., *Thematic Cartography and Remote Sensing*, Concept Pub. Co., New Delhi,1992.
9. Nelson Petrie*, Analysis and Interpretation of Topographical Maps (Rev),* OrientBlackswan Pvt. Ltd. 1997.
10. Robinson, A.H. et. al, *Elements of Cartography*, John Wiley & Sons, New York, 2009.

Further Readings:

1. Harley J.B. and David Woodward, *The History of Cartography* Volume 2, Book 2,The University of Chicago Press, 1995.
2. Raisz, Erwin, *Principles of Cartography*, McGraw Hill, New York, 1962.
3. Ramamurthy, K., *Maps Interpretation: India Landscapes Through Survey of India*,Topographic Maps, R.K. Mutt Road, Madras, 1982.
4. Rampal, K.K., *Mapping and Compilation*, Concept Publishing Co., New Delhi, 1993.
5. Tyner, J.A., *Principles of Map Design*, Guilford Publications, New York, 2010.
6. Wood Clifford H. and Keller C., *Cartographic Design- Theoretical and PracticalPerspectives*, John Wiley & Sons, 1996.

Pedagogy: Students will be encouraged to ask questions and participate in classroom discussions. Assignments, term papers and group discussions will be the additional methods to make teaching more focused and interesting.

GI 202: SPATIAL ANALYSIS

Max. Marks: 100
Theory: 50
Practical: 30
Internal Assessment: 20

Objectives: The course is designed to help the students to understand the tools and techniques which assist in providing focused and precise analysis of emerged pattern of geographical phenomena. The aim is to enhance the capabilities of students in perceiving, analysing and creating statistical information with sound geographical basis.

(A) THEORY **Marks: 50**

Unit- I

Fundamentals of Spatial Analysis

i. Concept and Significance of Spatial Analysis
ii. Nature of Spatial Data and Data Models
iii. Measures of Central Tendency: Mean, Median and Mode

Unit- II

Spatial Techniques for Point and Line Analysis

iv. Measurement of Point Features over Space
v. Measurement of Linear Features over Space
vi. Network Analysis

Unit-III

Vector Analysis Tools and Techniques

vii. Query Techniques; Feature Extraction Techniques
viii. Proximity Analysis Techniques; Overlay Techniques
ix. Statistical Techniques: Nearest Neighbour Analysis; Spatial Autocorrelation

Unit-IV

Raster Analysis: Elevation Models and Surface Analysis

x. Elevation Models: DEM, DTM, DSM and TIN
xi. Surface Analysis: Characterisitcs, Types and Representation
xii. Surface Creation: Interpolation Techniques; Raster Operations

Note:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Familiarization with Spatial Analysis Software.
2. Measuring Mean Centre, Weighted Mean Centre and Median Centre
3. Measuring Standard Distance and Standard Distance Ellipse
4. Nearest Neighbour Analysis
5. Measuring Spatial Autocorrelation
6. Network Analysis
7. Vector Analysis: Buffering, Overlay and Query
8. Interpolation Techniques and Analysis
9. Surface Analysis using DEM
10. Model Building in GIS

Note: The practical examination shall be conducted by a team of three examiners, including the faculty teaching the paper, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Chang, Kang-tsung, *Introduction to Geographic Information Systems,* 8th Edition,Tata McGraw-Hill Education, 2015.
2. Fotheringham Stewart A and Rogerson A. Peter, *The SAGE Handbook of Spatial Analysis*, SAGE Publications, London, 2009.
3. Laurini, R. & Thomson, D., *Fundamentals of Spatial Information Systems*, Academic Press, 1994.
4. Sanders, L., *Models in Spatial Analysis*, John Wiley & Sons, 2013.
5. Wong, David W.S. & Jay Lee, *Statistical Analysis of Geographic Information with ArcView GIS and ArcGIS*, John Wiley and Sons, 2005.

Further Readings:

1. Burrough, P.A. & Macdoneli, R.A., *Principles of Geographic Information Systems,* Oxford University Press, 2000.

2. Demers, M.N., *Fundamentals of Geographic Information Systems*, 2nd Edition, John Wiley and Sons, 2000.
3. Lloyd, Christopher D., *Spatial Data Analysis: An Introduction for GIS Users*, Oxford University Press Inc., New York, 2010.
4. Maguire, D.J.; M. Batty and M.F. Goodchild, *GIS, Spatial Analysis and Modeling*, ESRI Press, 2005.
5. Murayama, Y. and R. Thapa, *Spatial Analysis and Modeling in Geographical Transformation Process: GIS Based Applications*, Springer, 2011.

Pedagogy: Based on participatory teaching-learning, students shall be encouraged to enhance their skills through joint productive activity. Learning-by-doing shall be the guiding principle with a focus on specifically designed hands-on-training and group exercises. Students may be asked to prepare a report on outcomes of different spatial techniques. Students will be encouraged to discuss important themes amongst peers and seek their answer by participating in active classroom discussions

GI 203: DIGITAL IMAGE PROCESSING

Max. Marks : 100
Theory : 50
Practical : 30
Internal Assessment : 20

Objectives: The course is designed to introduce the student to analytical tools and methods which are currently used in digital image processing. The idea is to improve their capabilities for image information extraction by applying cutting edge tools available for satellite image restoration, enhancement and information extraction.

A. THEORY **Marks: 50**

Unit- I

Digital Image Pre-processing

i. Concept and Characteristics of Digital Image and Image Processing
ii. Radiometric Correction: Sources, Types and Correction Methods
iii. Geometric Correction: Sources, Types and Correction Methods.

Unit-II

Image Enhancement Techniques

iv. Introduction to Image Enhancement Techniques
v. Radiometric Enhancement Techniques
vi. Spatial Enhancement: Filtering Techniques

Unit-III

Image Transformation: Spectral Enhancement

vii. Image Arithmetic and Vegetation Indices
viii. Principal Component Analysis (PCA)
ix. Intensity, Hue and Saturation; Fourier Transformation

Unit- IV

Digital Image Classification

x. Image Classification Approaches and Techniques
xi. Supervised and Unsupervised Classification
xii. Accuracy Assessment; Change Detection and Analysis

Note:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25- 30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Familiarization with Image Processing Software
2. Georeferencing/Image Registration and Re-projection
3. Layer Stacking, Image Mosaic & Image Subset
4. Radiometric Enhancement Techniques
5. Applications of Filtering Techniques
6. Applications of Vegetation Indices
7. Principal Component Analysis (PCA)
8. Unsupervised Classification and Accuracy Assessment
9. Supervised Classification and Accuracy Assessment
10. Change Detection Analysis

Note: The practical examination shall be conducted by a team of three examiners, including the faculty teaching the paper, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Burger W., & Burge, M.J., *Principles of Digital Image Processing,* Springer-Verlag London Limited, 2009.
2. Campbell, J. B. and R.H. Wynne,*Introduction to Remote Sensing* (Fifth Edition), Guilford Press, 2012.
3. Chen, C.H., *Image Processing for Remote Sensing,* CRC Press, 2008.
4. Gonzalez, R.C. & Woods, R.E., *Digital Image Processing*, 2nd edition, Prentice-Hall, Inc., 2002.
5. Jensen, J.R., *Introductory Digital Image Processing: A Remote Sensing Perspective*, Prentice Hall, New Jersey, 1986.
6. Liu, J.G. & Mason, P.J., *Essential Image Processing and GIS for Remote Sensing,* John Wiley & Sons, 2009.
7. Mather, P.M. & Magaly, K., *Computer Processing of Remotely-Sensed Images: An Introduction,* Fourth Edition, John Wiley & Sons, 2011.
8. Richards, John A.,*Remote Sensing Digital Image Analysis: An Introduction,* 5th Edition, Springer Verlag Berlin Heidelberg, 2013.

Further Readings:

1. Dong, P. and Qi Chen, *LiDAR Remote Sensing and Applications*. CRC Press, Taylor & Francis Group, 2018.
2. Gomarasca, Mario A, *Basics of Geomatics*, Springer: Heidelberg, 2009.
3. Jain, A.K., *Fundamentals of Digital Image Processing,* Prentice-Hall, Inc., 2002.
4. Joseph, G. *Fundamentals of Remote Sensing*, Universities Press, 2005.
5. Lillesand, T.M., Kiefer, R.W. & Chipman, J. W., *Remote Sensing and Image Interpretation*, Fourth Edition, John Wiley and Sons, 2004.
6. Pratt, W.K., *Digital Image Processing*, 3rd Edition, John Wiley & Sons, 2001.
7. Prost, Gary L., *Remote Sensing for Geoscientists: Image Analysis and Interpretation*, 3rd Edition, CRC Press, Taylor & Francis, 2013.
8. Schowengerdt, Robert A. *Remote Sensing: Models and Methods for Image Processing*, Second Edition, Academic Press: San Diego, 2017.

PEDAGOGY: Learning shall be promoted through joint productive activity with a focus on participatory learning, hands-on-training and group exercises. Students may be asked to prepare a report of outcomes of image processing techniques applied on satellite images. Students will be encouraged to ask questions and participate in classroom discussions.

GI 204: FUNDAMENTALS OF GLOBAL POSITIONING SYSTEM

Max. Marks : 100
Theory : 50
Practical : 30
Internal Assessment : 20

Objectives: The objectives of this paper are to give a comprehensive introduction to GPS and DGPS technology, system concepts, design, operation, implementation and applications, to provide detailed information on the GPS signal, the processing by the receiver, and the techniques by which GPS obtains position, velocity and time.

(A) THEORY **Marks: 50**

Unit-I

i. Concept and Characteristics of Global Navigational Satellite System (GNSS)
ii. Historical Background, GPS Segments, Features of GPS Satellites, Principles ofOperation, Fundamentals on Satellite Geodesy, Co-ordinate Systems related to GNSS,Survey and Characteristics of Observed Values.

Unit-II

iii. Surveying with GPS: Absolute, Relative and Differential Positioning, Kinematic GPS.
iv. Performing GNSS Measuring and Data Processing, Overview on Geodetic Dataand Data Transformation

Unit-III

v. Survey on GNSS Application: Basic Principles in Working with DGNSS
vi. Methods for Determining and Techniques for Searching Ambiguities both for Phase Data and the Combination of Code and Phase Data; GNSS Application in: Geodesy, Geodynamics, Tectonic Plate Movement with Time Alternating Coordinates and Navigation.

Unit-IV

vii. Diverse Systems for Global Positioning: GPS, GLONASS, GALILEO, Beidou, IRNSS
viii. Location-based Services: Practical Application.

Note:

1. A Compulsory Question at serial number I containing 10 short answer type questionsshall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from eachunit. The candidates will attempt 4 questions selecting one from each unit carrying 9marks each (total 36 marks). These will be in addition to the Compulsory Question atserial number I.

(B) PRACTICAL **30 Marks**

1. Components and Operation of GPS Instrument
2. Understanding the Different Pages of GPS
3. Collecting Readings using Primary Pages
4. Surveying a Small Area
5. Mapping of the Surveyed Area
6. Tabulation of Information of the Surveyed Area
7. Analysis of the Map and Data Collected
8. Application Exercises

Note: The practical examination shall be conducted by a team of three examiners, including the faculty teaching the paper, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Anderle, R., *The Global Positioning System*, Royal Society of London, UK, 1988.
2. Blewitt, G., *Advances in GPS Technology for Geodynamics Investigations: 1978-1992*, Crustal Dynamics Project AGU Monograph, NASA. 1993.
3. Colombo O. L. & M. M. Watkins, *Satellite Positioning*, in US National report to IUGC on Geodesy, 1991.
4. El-Rabbani, Ahmed, *Introduction to GPS: The Global Positioning System*, Artech House, 2002.
5. Jones, *Geographical Information Systems and Computer Cartography*, Pearson Education Inc, 1997.
6. Kaplan, E. D. and C.J. Hegarty (eds.), *Understanding GPS/GNSS: Principles and Applications*, 3rd Edition, Artech House, 2017.
7. Leick, A, Lev Rapoport and Dmitry Tatarnikov, *GPS Satellite Surveying*, 4th Edition, John Wiley & Sons, 2015.
8. Letham, Lawrence, *GPS Made Easy: Using Global Positioning Systems in the Outdoors*, Rocky Mountain Books, 2008.
9. Rao, G.S., *Global Navigation Satellite Systems- With Essentials of Satellite Communications*, Tata McGraw Hill Education Private Limited, New Delhi, 2010.

Further Readings:

1. Fell, P. J., *Geodetic Positioning Using GPS*, The Ohio State University Report no. 299, Columbus, Ohio, USA, 1980.
2. Goad, C. C., *Proceedings of First International Symposium on GPS*, USA, 1985.
3. Gomarasca, Mario A, *Basics of Geomatics*, Springer: Heidelberg, 2009
4. King, R. W. & Others, *Surveying GPS*, University of New South Wales, Australia, 1985.
5. Noureldin. A., Karamat, T.B. and Jacques Georgy, *Fundamentals of Inertial Navigation, Satellite-based Positioning and their Integration*, Springer, 2013.
6. Rizos, *Introduction to GPS*, University of New South Wales, 1999.
7. Srivastava, G.S., *An Introduction to Geoinformatics*, McGraw Hill Education, India, New Delhi, 2014

PEDAGOGY: Students will be provided demo exercises to increase their awareness about wide ranging applications of GPS. Students may be asked to prepare a database of various themes gathered using GPS devices and analyse the data after relevant post-processing. Students will be encouraged to ask questions and participate in classroom discussions.

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